

Partnerships Between Cloud Service Providers and AI Developers

FTC Staff Report on AI Partnerships & Investments 6(b) Study



Federal Trade Commission
Office of Technology Staff
January 2025

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1. Executive Summary

This report presents FTC Office of Technology (“OT”) staff’s findings from the agency’s 6(b) study into three partnerships involving generative artificial intelligence (“AI”): Microsoft-OpenAI, Amazon-Anthropic, and Google-Anthropic.¹ These partnerships involve relationships between the world’s current largest Cloud Service Providers (“CSPs”) and two of the most prominent AI model developers. These partnerships therefore have potential for significant impact on AI technology, workers, and consumers. Through the 6(b) study, OT staff sought to better understand the nature of these relationships, including any ongoing obligations between the companies, to evaluate their potential impact on the marketplace.²

For an individual using an AI chatbot to write a wedding speech or a small business owner generating logo ideas with an AI image generation model, corporate partnerships may seem abstract or distant. Yet, these partnerships may potentially impact AI model development—including which firms may effectively participate in the marketplace—and may determine many aspects of those individuals’ and firms’ experiences. For these reasons, the agency’s 6(b) study sought to understand the motivations and impacts of these partnerships on cloud computing and generative AI spaces.

The Commission issued the 6(b) orders to the five respondents in January 2024 and the respondents provided information through September 2024. AI technology is rapidly evolving, and therefore the context of this work may continue to evolve. The findings represent the information available to OT staff about these corporate partnerships as of September 2024 as well as publicly available information through January 2025.

This report is aimed at helping the agency, the public, and policymakers deepen their understanding of the corporate partnerships formed between the generative AI developers and CSPs included in this study. This report summarizes the findings from the 6(b) study, providing timely insight to the general public that is not otherwise available, including through previous reporting on the partnerships. It also equips the agency to better evaluate the attributes and potential implications of partnerships involving large technology companies and the impact they may have on hundreds of millions of consumers, countless businesses, and huge segments of the economy now, and not years after the fact. Even in places where public reporting had already described certain partnership features, non-public information helped to enrich staff’s—and therefore this report’s—analysis of the partnerships.

The information presented in this report has been aggregated or anonymized as necessary to protect trade secrets and confidential or privileged commercial or financial information submitted by respondents.

Staff notes that this report is not a formal legal or economic analysis of the partnerships or markets and should not be interpreted as such. The observations discussed in this report are limited to a narrow set of information and documents submitted by the five 6(b) respondents through September 2024. A comprehensive analysis of the subject partnerships is beyond the scope of the report. The technologies

¹ For the purposes of this report, generative AI is used to refer to the class of machine learning models that emulate the structure and characteristics of input data in order to generate derived synthetic content. This can include images, videos, audio, text, and other digital content, as defined in Exec. Order No. 14110, 88 Fed. Reg. 75191 (Nov. 1, 2023), <https://www.federalregister.gov/documents/2023/11/01/2023-24283/safe-secure-and-trustworthy-development-and-use-of-artificial-intelligence>. This report uses the term AI as a shorthand for generative AI, because generative models are the only ones we discuss here except where otherwise noted.

² In this report we use the terms “marketplace” and “market” interchangeably. The use of these terms is meant to reference the word as it is generally understood; it should not be construed as an attempt to define an antitrust market.

discussed in this report are rapidly evolving, and therefore the implications of the partnerships may continue to evolve.

Summary of Findings

The report contributes to the public’s understanding of the evolving AI landscape in three ways. After providing background and summarizing the context in which this study was conducted, the report first describes the AI technology stack, according to OT staff’s current understanding. Second, the report summarizes key terms of the AI partnerships based on review of the non-public documents produced in response to the study, as well as public reporting on, and public statements by, the respondents. Third, the report highlights certain areas to watch going forward: how the AI partnerships may affect access to certain inputs, how contractual terms and technical barriers may affect switching costs, and whether information exchanged in the partnerships might provide CSP partners access to sensitive technical and business information that may be unavailable to others. Since the study was limited to three corporate partnerships, any analysis of partnership terms in this report should not necessarily be construed to apply to all potential relationships between CSPs and AI developers.

Through information received via the 6(b) study, including non-public information, staff explored the potential “nature and extent of interactions with and influence over” each respondent from the partnership.³

Key Terms of the AI Partnerships

Section 4 of this report highlights the following key terms⁴ of the AI partnerships:

1. **The partnerships provide CSP partners significant equity and certain revenue-sharing rights in their AI developer partners.** The partnerships also leave open the possibility that the AI developers could be fully acquired by the CSPs in the future.
2. **The partnerships provide CSP partners certain consultation, control, and exclusivity rights with respect to their AI developer partners to varying degrees.** In some cases, the agreements include exclusive or preferential treatment of the CSP partner and the ability to potentially influence the AI model developer partner through board seats and consultation rights.
3. **The partnerships include cloud commitments that require AI developers to spend a large portion of their CSP partner’s investment on cloud services from their partner.** This feature of the partnerships—circular spending—is one avenue through which CSP partners may potentially aim to reduce the magnitude of potential loss invested directly into their partners and cloud infrastructure to serve their partners.
4. **The partnerships permit the sharing of key resources and information.** These resources include, to varying degrees, discounted access to computing resources for AI developer partners; intellectual property, and performance and financial data for CSP partners; and engineering personnel and training data for both partners. The partnerships include co-development plans for CSP-designed semiconductor chips optimized for the needs of AI developer partners’ models.

³ Fed. Trade Comm’n, AI Investments and Partnerships 6(B) Study Sample Order (2024), https://www.ftc.gov/system/files/ftc_gov/pdf/P246201_AI_Investments_6%28b%29_Order_and_Resolution.pdf.

⁴ Each of these terms may not apply to each of the partnerships; more specificity can be found in Section 4 of the report.

5. **The partnerships offer CSP and AI developer partners opportunities to expand current products.** In some cases, this includes integration into CSP partners' first-party products or deployment of AI developer partners' models on CSP partners' cloud platforms.

Areas to Watch Regarding Potential Implications of the AI Partnerships

Section 5 of this report identifies certain areas to watch going forward, which are not necessarily representative of all the partnerships' potential implications. A fulsome analysis of these issues is beyond the scope of this report but could be done using traditional antitrust tools and principles. The areas to watch include the following:

1. **The partnerships could affect access to certain inputs, such as computing resources and engineering talent.** The partnerships might affect the likelihood that a CSP partner would consider limiting access to inputs such as computing resources for AI developers other than its partner. The partnerships could also affect the availability of AI developer engineering talent. This could impact competition for both AI developer partners and non-partner AI developers.
2. **The partnerships could increase contractual and technical switching costs for AI developer partners.** The partnership agreements may increase switching costs via contractual commitments (including exclusivity requirements) and technical barriers. This can make it more difficult for AI developer partners to change CSPs or restrict their use of multiple CSPs.
3. **The partnerships provide CSP partners access to sensitive technical and business information that may be unavailable to others.** The partnership agreements provide CSPs access to certain technical and business information that other CSPs and AI developers may not have, including information related to generative AI models, AI development methods, confidential chip co-design, partner finances, and customer usage and revenue numbers. CSPs may be able to use this information to develop their own internal models and applications, including products that may compete with those of their partners.

2. Background

2.1 - Current Technological Landscape

The rapid and widespread development and deployment of generative AI has driven significant changes across society, including knowledge acquisition, creativity, and productivity. Generative AI has captured interest across commercial sectors—including in banking and financial services, healthcare, transportation, legal and professional services, customer service, travel, journalism, education, and marketing.

Although commercial generative AI is not completely new, public interest and investments in generative AI have sharply increased in recent years.⁵ In 2023, billion-dollar AI partnerships began to form or grow, with the potential to impact the competitive dynamics of the space and therefore impact the quality and availability of AI products.

⁵ *Generative AI Venture Capital Investment Globally on Track to Reach \$12 Billion in 2024, Following Breakout Year in 2023*, EY IRELAND (May 16, 2024), https://www.ey.com/en_ie/news/2024/05/generative-ai-venture-capital-investment-globally-on-track-to-reach-12-billion-dollar-in-2024-following-breakout-year-in-2023.

4.1 - The partnerships provide CSP partners significant equity and certain revenue-sharing rights in their AI developer partners.

As a result of the CSPs' ownership interests in, or rights to profits from, the AI developer partner, the CSP partners might benefit financially if their AI developer partner's value increases.¹²³ Staff notes that profit allocated to Microsoft by OpenAI is reportedly currently capped at \$92 billion.¹²⁴

Public reporting has indicated that both Amazon and Google have non-voting shares in Anthropic.¹²⁵ Staff is currently not aware of a public disclosure regarding the nature of Microsoft's ownership in OpenAI.

Public reporting has also posited that each of the partnerships contain revenue-sharing agreements that provide income for CSPs based on the success of their partner's AI models, including sharing revenue from the model developer partner's own first-party API.¹²⁶ Internal documents confirmed that revenue sharing is part of multiple partnerships in various ways, including [REDACTED]

The 6(b) documents reviewed by staff demonstrated that at least some respondents recognized that the acquisition of the AI developer by its CSP partner was a possibility.¹²⁸ One respondent, for example, wrote in internal documents that [REDACTED]

Other documents also indicated that some respondents were less receptive to a CSP acquiring its AI developer partner.¹³⁰ AI developer partners and CSP partners have also considered extending their partnerships with further funding.¹³¹

¹²³ See, e.g., Respondent Document Submission [REDACTED] for one company's discussion of potential investment outcomes.

¹²⁴ Our structure, OPENAI, <https://openai.com/our-structure/>; Jessica Mathews and Jeremy Kahn, *Inside the Structure of OpenAI's looming new investment from Microsoft and VCs*, FORTUNE (Jan. 11, 2023), <https://fortune.com/2023/01/11/structure-openai-investment-microsoft/>.

¹²⁵ Billy Perrigo, *How Anthropic Designed Itself to Avoid OpenAI's Mistakes*, TIME (May 30, 2024), <https://time.com/6983420/anthropic-structure-openai-incentives/>.

¹²⁶ Anissa Gardizy, *Amazon Discussing New Multibillion-Dollar Investment in Anthropic*, THE INFORMATION (November 7, 2024), <https://www.theinformation.com/articles/amazon-discussing-new-multibillion-dollar-investment-in-anthropic>; see also Anissa Gardizy et al., *OpenAI Leaders Say Microsoft Isn't Moving Fast Enough to Supply Servers*, THE INFORMATION (Oct. 8, 2024), <https://www.theinformation.com/articles/openai-eases-away-from-microsoft-data-centers>.

¹²⁷ Respondent Document Submission [REDACTED]

¹²⁸ Respondent Document Submission [REDACTED]

¹²⁹ Respondent Document Submission [REDACTED]

¹³⁰ Respondent Document Submission [REDACTED]

¹³¹ See, e.g., Respondent Document Submission [REDACTED]

4.2 - The partnerships provide CSP partners certain consultation, control, and exclusivity rights with respect to their AI developer partners to varying degrees.

At least one partnership provides the opportunity for CSP executive leadership to consult with or advise their partner companies.¹³² Internal documents described that one partnership gave the CSP partner input on [REDACTED]¹³³ and at least one partnership also gave the CSP partner advance notice of important decisions.¹³⁴ These rights may provide a potential way for CSP partners to influence the AI model developer partners. However, these rights vary in the extent to which they have been exercised and the potential they have for impacting the day-to-day operations of AI developer partners.

Control can also come in the form of who is on the company's board of directors. Board members have several key duties which vary from organization to organization, but generally include strategic oversight, planning, budget approval and fundraising, legal compliance, and decision-making on significant corporate changes.¹³⁵

For example, Microsoft previously occupied an observer seat on the board of OpenAI.¹³⁶ While it relinquished that seat in 2024, a company spokesperson also noted that Microsoft will continue to receive information from OpenAI and that they will "host regular stakeholder meetings to share progress on our mission and ensure stronger collaboration across safety and security."¹³⁷ In addition, one board member served on both Microsoft's and OpenAI's board contemporaneously for over three years.¹³⁸

CSP partners also have certain exclusivity or parity rights through the partnerships—such as access to certain products, distribution capabilities, or brand association.

¹³² See, e.g., Respondent Document Submission [REDACTED]

¹³³ Respondent Document Submission [REDACTED]

¹³⁴ [REDACTED]

¹³⁵ See, e.g., Kay Brancato et al., *The Role of U.S. Corporate Boards in Enterprise Risk Management*, THE CONFERENCE BOARD RESEARCH REPORT NO. R-1390-06-RR (Nov. 1, 2006), <https://www.conference-board.org/publications/publicationdetail.cfm?publicationid=1190>.

¹³⁶ Foo Yun Chee, *Microsoft ditches OpenAI board observer seat to stave off antitrust scrutiny*, REUTERS (July 10, 2024), <https://www.reuters.com/technology/microsoft-ditches-openai-board-observer-seat-amid-regulatory-scrutiny-2024-07-10/>.

¹³⁷ Mauro Orru and Christian Moess Laursen, *Microsoft Quits OpenAI's Board Amid Antitrust Scrutiny*, WALL STREET JOURNAL (July 10, 2024), <https://www.wsj.com/tech/ai/microsoft-withdraws-from-openais-board-amid-antitrust-scrutiny-aab6ff1e>.

¹³⁸ Kif Leswing, *Reid Hoffman steps down from OpenAI board to avoid potential conflicts of interest*, CNBC (Mar. 3, 2023), <https://www.cnbc.com/2023/03/03/reid-hoffman-steps-down-from-openai-board-to-avoid-potential-conflicts.html>.

In some cases, the agreements include exclusive or preferential treatment to the CSP partner,¹³⁹ or require parity with other cloud partners or the AI developer's own products.¹⁴⁰ Staff also notes that although the existence of exclusivity, preference, or parity provisions varied by partnership, in general, multiple partnerships have provisions that impose conditions or restrictions on the ability of AI developer partners to share accrued benefits with other non-partner cloud providers or companies.¹⁴¹

One CSP described the different ways in which its partnership could potentially provide parity as follows:

"We've sought parity in multiple ways: Parity of making the models and updates of the same available at the same time as they do [redacted] [core ask]. Parity on features and functionality including performance, reliability, support, terms of service, regional availability, etc. Parity on pricing [redacted] vs [redacted] (they set the pricing). Parity on compensation for their sellers on sales on [redacted] (as compared to [redacted]). Parity on marketing (promotion of [redacted] as with [redacted])."¹⁴²

In at least one case, a partnership effectively required the AI developer partner to keep the models on its partner's cloud service at parity with the models on the AI developer's first-party API.¹⁴³ In other words, the AI developer would not be able to release a new model on its own before launching it on its partner's cloud service. Nor would the AI developer partner be able to offer substantially better models to business or individual consumers as an incentive to use its models through a first-party API rather than through its partner's cloud.

4.3 - The partnerships include cloud commitments that require AI developers to spend a large portion of their CSP partner's investment on cloud services from their partners.

According to public reporting summarized in Table 2, the AI developer partners committed to spend billions of dollars on cloud services from their partners.¹⁴⁴ The committed spending on cloud computing means that, to a significant extent, the CSP is effectively investing in the partnership through the provision of cloud computing. This feature of the partnerships—circular spending—is one avenue

¹³⁹ See, e.g., Respondent Document Submission [redacted].

¹⁴⁰ See, e.g., Respondent Document Submission [redacted].

¹⁴¹ See [redacted].

See also [redacted].

Respondent Document Submission [redacted].

¹⁴² Respondent Document Submission [redacted].

¹⁴³ See, e.g., Respondent Document Submission [redacted].

[redacted], stating that [redacted].

See also Respondent Document Submission [redacted].

[redacted], citing [redacted].

See Respondent Document Submission [redacted].

¹⁴⁴ See Jin & Dotan, *supra* note 7.

through which CSP partners may have potentially aimed to reduce the magnitude of potential loss invested directly into their partners and cloud infrastructure to serve their partners.

Table 2

Summary of Publicly Reported Spending Commitment \$			
Cloud spending commitment by AI developer	Microsoft-OpenAI	Amazon-Anthropic	Google-Anthropic
	“Billions” ¹⁴⁵	At least \$4 billion, not including commit after latest \$4 billion investment ¹⁴⁶	At least \$3 billion ¹⁴⁷

AI developers’ committed funds requirements varied by partnership, but funds can be spent on a range of computational tasks needed by AI developers, including training or fine-tuning foundation models; inference, *i.e.*, computing responses to consumer prompts on the AI developers’ own products or via CSP model platforms; or storage services, such as storing training data or prompts.¹⁴⁸

For Microsoft and OpenAI specifically, according to publicly disclosed information, OpenAI has been using Microsoft as its “exclusive cloud provider” dating since the start of their partnership in 2019, which was reaffirmed in 2023.¹⁴⁹

At least one CSP order recipient claimed publicly that cloud spending commitments are separate from investment deals, and explained, for example, that the decision to invest in its developer partner was made by a separate team than the team negotiating any cloud spending commitment agreements.¹⁵⁰ However, 6(b) documents suggest that multiple respondents may view investments as linked to cloud spending commitments.¹⁵¹ For example, in a list of strategic rationales for the partnership, one CSP listed first that the partnerships provided a “[l]arge [cloud service] [redacted] commit opportunity.”¹⁵²

One CSP noted that the cloud spend commitment had limited downside since any AI infrastructure built for its AI partner could be resold or used internally: “If [redacted] cannot raise/generate more funds, we

¹⁴⁵ See Jin & Dotan, *supra* note 7.

¹⁴⁶ *Id.*; See also Amazon Staff, *supra* note 9.

¹⁴⁷ *Id.*

¹⁴⁸ Respondent Document Submission [redacted] Respondent Document Submission [redacted]

¹⁴⁹ *Microsoft invests in and partners with OpenAI to support us building beneficial AGI*, OPENAI (July 22, 2019), <https://openai.com/index/microsoft-invests-in-and-partners-with-openai/> (“We’ll jointly develop new Azure AI supercomputing technologies, and Microsoft will become our exclusive cloud provider—so we’ll be working hard together to further extend Microsoft Azure’s capabilities in large-scale AI systems.”); *Microsoft and OpenAI extend partnership*, MICROSOFT (Jan. 23, 2023), <https://blogs.microsoft.com/blog/2023/01/23/microsoftandopenaiextendpartnership/> (“Exclusive cloud provider - As OpenAI’s exclusive cloud provider, Azure will power all OpenAI workloads across research, products and API services.”).

¹⁵⁰ See Jin & Dotan, *supra* note 7.

¹⁵¹ Respondent Document Submission [redacted] Respondent Document Submission [redacted]

¹⁵² Respondent Document Submission [redacted]

As staff has laid out above, generative AI requires extensive resources and funding to provide access to infrastructure, access to talent, and access to data to train foundation models. This is why some firms have used technology partnerships, investments, and other deals to accelerate development and deployment of AI models. While these types of deals are not a new phenomenon, they have seen a high-profile resurgence due to the generative AI wave and increased merger scrutiny.²⁶³ The partnerships examined in the present report are prime examples of this resurgence—making it important that the FTC and the public understand the terms of the partnerships and their impacts on competition and consumers.

Through a scoped and targeted 6(b) process, staff were able to obtain core non-public pieces of information regarding partnership terms and establish areas to watch going forward related to potential impacts of the partnerships on the AI ecosystem.

²⁶³ See, e.g., Press Release, Fed. Trade Comm’n, FTC and DOJ Seek Info on Serial Acquisitions, Roll-Up Strategies Across U.S. Economy (May 23, 2024), <https://www.ftc.gov/news-events/news/press-releases/2024/05/ftc-doj-seek-info-serial-acquisitions-roll-strategies-across-us-economy>.